



Calibrating Your Display on a Budget

Quick and Easy Display Calibration with Built-in Settings and a White Sheet of Paper

While professional calibration tools offer the highest accuracy, you can still achieve decent results using just your display's built-in settings and a white sheet of paper. This method won't be as precise as using a colourimeter, but it's a quick and cost-effective solution for those who want to improve their display's performance without investing in additional tools. Here's a simple step-by-step guide to calibrate your display with minimal resources.

Step 1: Prepare Your Environment

1. Find a room with consistent, diffused lighting, avoiding direct sunlight or bright light sources.
2. Turn on your display and let it run for at least 30 minutes to warm up.
3. Reset your display to factory settings.

Step 2: Calibrate Brightness

1. Open an image with a wide range of greys, such as a greyscale gradient or a photograph with shadow details.
2. Hold the white sheet of paper next to your display or monitor.
3. Adjust your display's brightness setting so that the white on your display matches the brightness of the white paper under the room lighting.

Step 3: Calibrate Contrast

1. Open an image with a wide range of black and white tones, such as a black-to-white gradient or a high-contrast photograph.
2. Adjust the contrast setting on your display to achieve the maximum distinction between the darkest black and the brightest white without losing detail in either extreme.

Step 4: Calibrate Colour Temperature

1. Place the white sheet of paper in front of your display, covering a portion of the screen.
2. Open an image with a neutral grey background or a white-to-black gradient.
3. Compare the white or grey on your display to the white paper.
4. Adjust the colour temperature setting (usually found under 'colour settings', 'colour balance', or 'white balance') to match the white or grey on your display to the white paper as closely as possible. You may need to experiment with different presets or manual adjustments to achieve the best result.

Step 5: Fine-tune Colour Balance (Optional)

If your display offers individual colour channel adjustments, you can further fine-tune the colour balance:

1. Open an image with a wide range of colours, such as a colour wheel or a photograph with various hues.
2. Compare the colours on your display to the same colours in real life or reference images on a known accurate display.
3. Adjust the individual red, green, and blue colour channels to achieve the closest match to the reference colours.

A Simple Solution for Display Calibration

While this method is not as accurate as using a colourimeter and calibration software, it can provide a noticeable improvement in your display's performance. By following these simple steps, you'll be able to achieve a more balanced and accurate visual experience using just your display's built-in settings and a white sheet of paper.

sources that could interfere with the colourimeter readings. Ideally, work in a room with consistent, diffused lighting.

2. **Display warm-up:** Turn on your display and let it run for at least 30 minutes before calibration to ensure consistent temperature

and colour performance.

3. **Display settings:** Reset your display to factory settings to start with a clean slate. Disable any auto-brightness or dynamic contrast features, as these can interfere with the calibration process and might skew the result.

Calibration in Action: A Step-by-Step Guide

Step 1: Install and launch the calibration software

Install your chosen software and launch it. If you're using a proprietary software that comes with your col-